

Communication Laboratory

Experiment 2: SWITCHING MODULATOR



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1. Introduction:

A switching modulator is a practical implementation of amplitude modulation based on the principle of signal multiplication using a high-speed electronic switch. It operates by periodically turning the message signal ON and OFF using a carrier signal, typically a square wave.

In this experiment, a JFET (BFW11) is used as the switching device. The carrier drives the gate terminal such that the JFET alternates between cutoff and triode region. This switching action effectively multiplies the sinusoidal message signal with a square wave, generating a Double Sideband Suppressed Carrier (DSB-SC) signal.

Since switching introduces higher-order harmonics, a second-order multiple-feedback Band-Pass Filter (BPF) is used to isolate the desired frequency band and obtain a clean amplitude-modulated signal.

2. Key Objectives

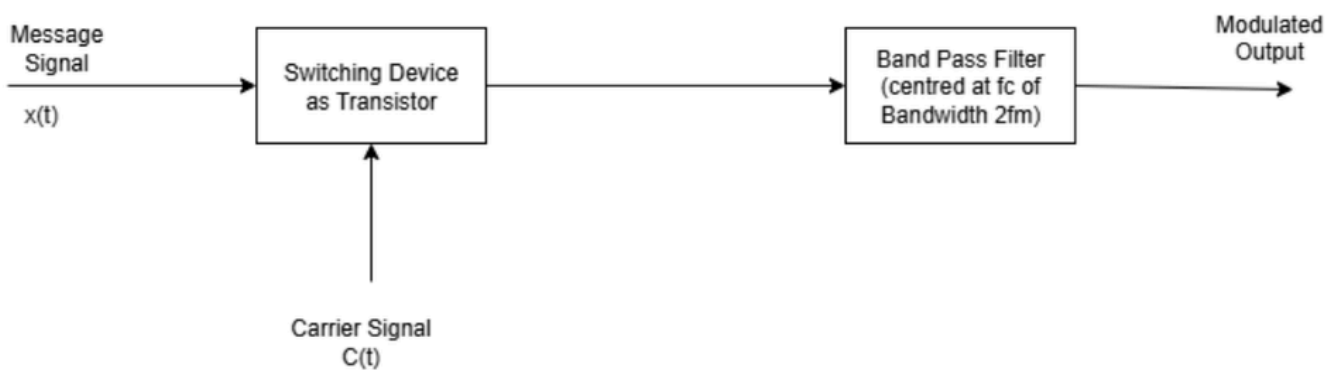
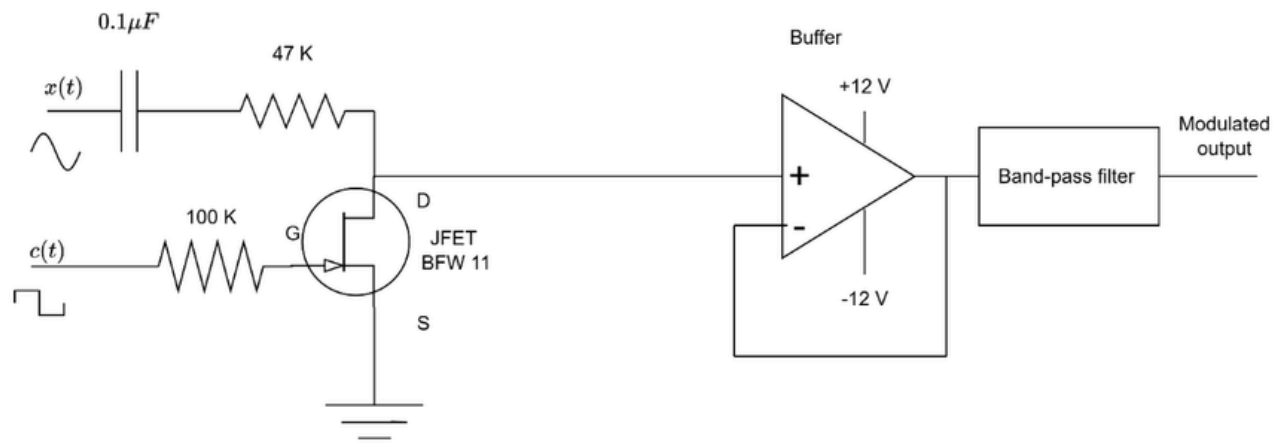


Figure 1 : Switching Modulator Using A Transistor

- To understand the principle of amplitude modulation using switching action.
- To design and implement a switching modulator for DSB-SC generation.



$x(t)$ = Message signal (Sine wave, $4V_{pp}$, $f_m=1\text{KHz}$),
 $c(t)$ = Carrier signal (Square wave, Duty cycle= 50%, $f_c=10\text{KHz}$)

Figure 2 : Circuit Diagram

- To design a second-order multiple feedback Band-Pass Filter (BPF).

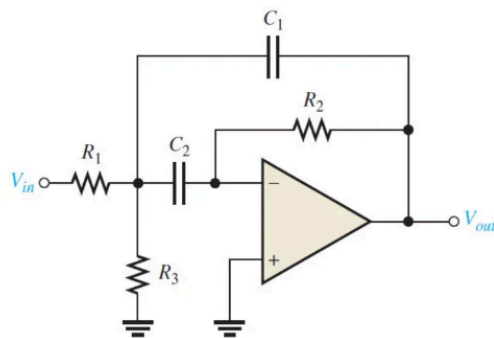


Figure 3: 2nd Order Band Pass Filter

To get the BPF Transfer Function we solve-

$$\frac{V_x - V_{in}}{R_1} + V_x(sC_2) + \frac{V_x}{R_3} + (V_x - V_{out})sC_1 = 0$$

$$V_x \left(\frac{1}{R_1} + sC_2 + \frac{1}{R_3} + sC_1 \right) = \frac{V_{in}}{R_1} + V_{out} sC_1$$

Solving, we get-

$$H(s) = \frac{\frac{s}{R_1 C_2}}{s^2 + s \left[\frac{1}{R_2 C_2} + \frac{1}{R_1 C_2} + \frac{1}{R_1 C_1} + \frac{1}{R_2 C_1} \right] + \frac{1}{R_2 R_1 C_1 C_2} + \frac{1}{R_1 R_L C_1 C_2}}$$

Comparing (3) with -

$$H(s) = \frac{Ks}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

We get-

$$f_c = \frac{1}{2\pi \sqrt{(R_1 \parallel R_3) R_2 C_1 C_2}}$$

$$Q = \frac{f_0}{BW}$$

Putting $C_1=C_2=C$ and solving R_1 , R_2 and R_3 We get-

$$R_1 = \frac{Q}{2\pi f_0 C A_0}, \quad R_2 = \frac{Q}{\pi f_0 C}, \quad R_3 = \frac{Q}{2\pi f_0 C (2Q^2 - A_0)}, \quad BW = \frac{1}{\pi R_2 C}$$

As $Q = 5$, $A_0 = 2$, Taking $C = 5.6\text{nF}$

We will get the values of R_1 , R_2 , R_3 , BW as follows :

$R_1 = 7.1\text{k}\Omega$, $R_2 = 28\text{k}\Omega$, $R_3 = 300\Omega$, $BW = 2\text{kHz}$

- To calculate mid-frequency, bandwidth, gain, and quality factor (Q).
- To observe time-domain and frequency-domain characteristics using CRO and FFT.

3. Components Used

- Op-Amp IC 741 (×2)
- Diode / JFET
- Resistors (7.1KΩ, 300Ω, 28KΩ)
- Capacitors (5.6 nF)
- Signal Generator
- Digital Storage Oscilloscope
- Breadboard

- DC Supply (+12V, -12V, 0V)
- Digital Multimeter

4. Theory

4.1 Amplitude Modulation

Amplitude Modulation is defined as the process in which the amplitude of the carrier wave $c(t)$ is varied linearly with the message signal $m(t)$.

$$\begin{aligned} x(t) &= \text{Message signal (Sine wave, } 4V_{pp}, f_m=1\text{KHz}), \\ c(t) &= \text{Carrier signal (Square wave, Duty cycle= 50\%, } f_c=10\text{KHz)} \end{aligned}$$

In a switching modulator, instead of directly multiplying signals using analog multipliers, a high-frequency carrier drives a switching device that alternately allows and blocks the message signal.

4.2 Switching Modulator

The switching modulator consists of:

1. Adder circuit (to combine carrier and message)
2. Switching element (Diode/JFET)
3. Band-Pass Filter

The carrier is usually a square wave. A square wave can be expressed using Fourier Series as:

$$p(t) = \frac{4}{\pi} \left[\cos(\omega_c t) + \frac{1}{3} \cos(3\omega_c t) + \frac{1}{5} \cos(5\omega_c t) + \dots \right]$$

Multiplying the message signal $m(t)$ with this switching function shifts the message spectrum to:

$$f_c \pm f_m, 3f_c \pm f_m, 5f_c \pm f_m, \dots$$

Thus, unwanted harmonics are generated along with the desired DSB-SC signal.

A Band-Pass Filter is therefore required to extract only the component around f_c .

4.3 JFET Switching Operation

- A negative DC offset is applied to the square wave to ensure negative V_{GS} , required to turn ON the JFET.
- When ON: $V_{DS} \approx 0V$, message passes.

- When OFF: High impedance, message blocked.

Care must be taken not to apply positive gate voltage to the BFW11 JFET, as forward biasing the gate-substrate junction may cause excessive current and device damage.

4.4 Band-Pass Filter Design

A second-order multiple-feedback band-pass filter is used.

Design specifications:

- Center frequency ≈ 10 kHz
- Bandwidth = 2 kHz (9 kHz – 11 kHz)
- Gain = 2

Quality factor:

$$Q = \frac{f_0}{BW}$$

$$Q = 5$$

A second-order filter is chosen because a first-order filter has insufficient roll-off to suppress harmonics effectively.

5. Results and Observations

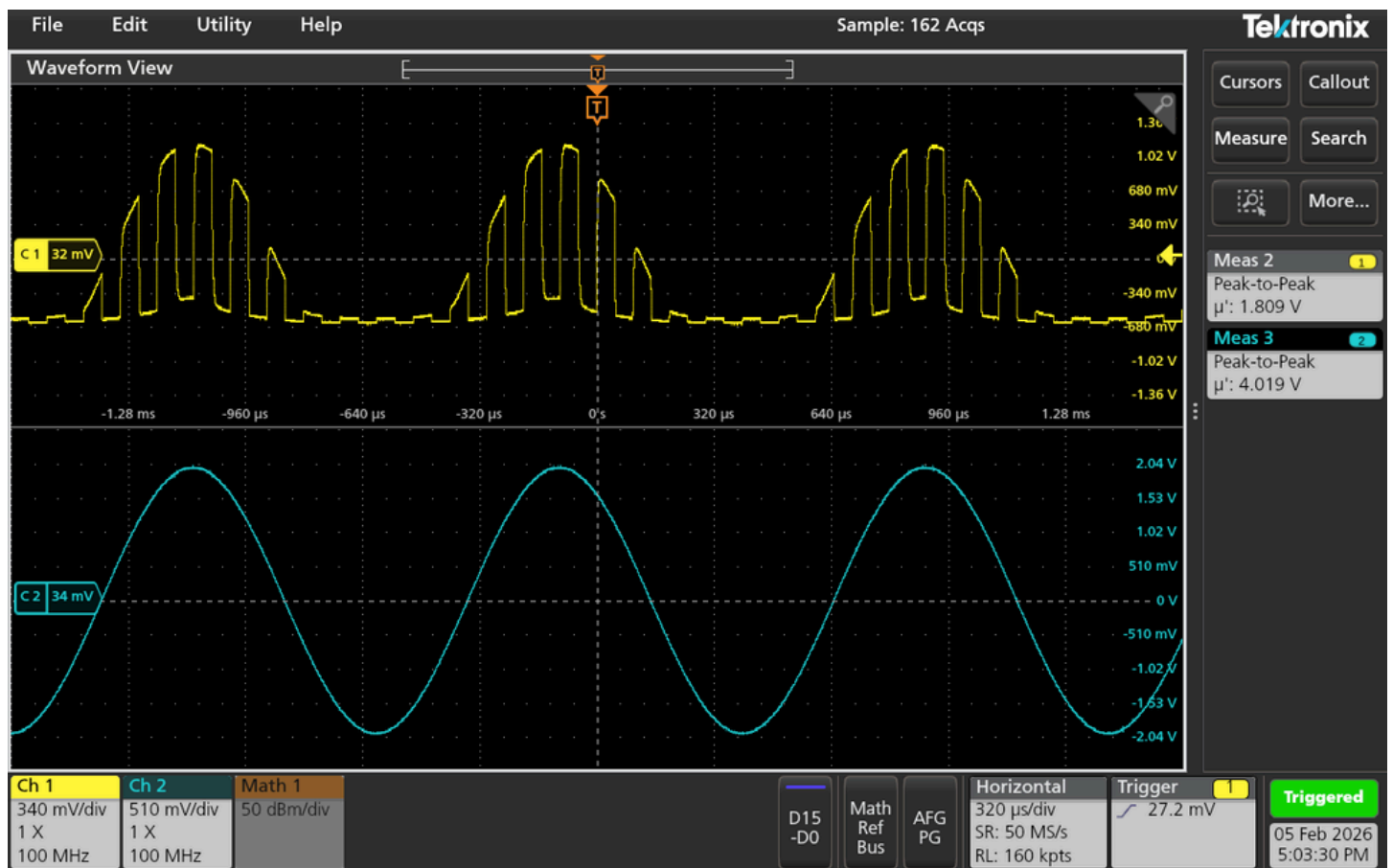


Figure 4: Output after MOSFET/JFET

At the drain of the BFW11 JFET:

The sine wave is periodically switched ON and OFF.

The waveform resembles a chopped sinusoid.

The envelope follows the message signal.

High-frequency carrier components are clearly visible.



Figure 5: Output after Op-Amp

The op-amp stage provides:

Signal amplification (Gain ≈ 2)

Improved signal level for filtering

Reduced loading effect

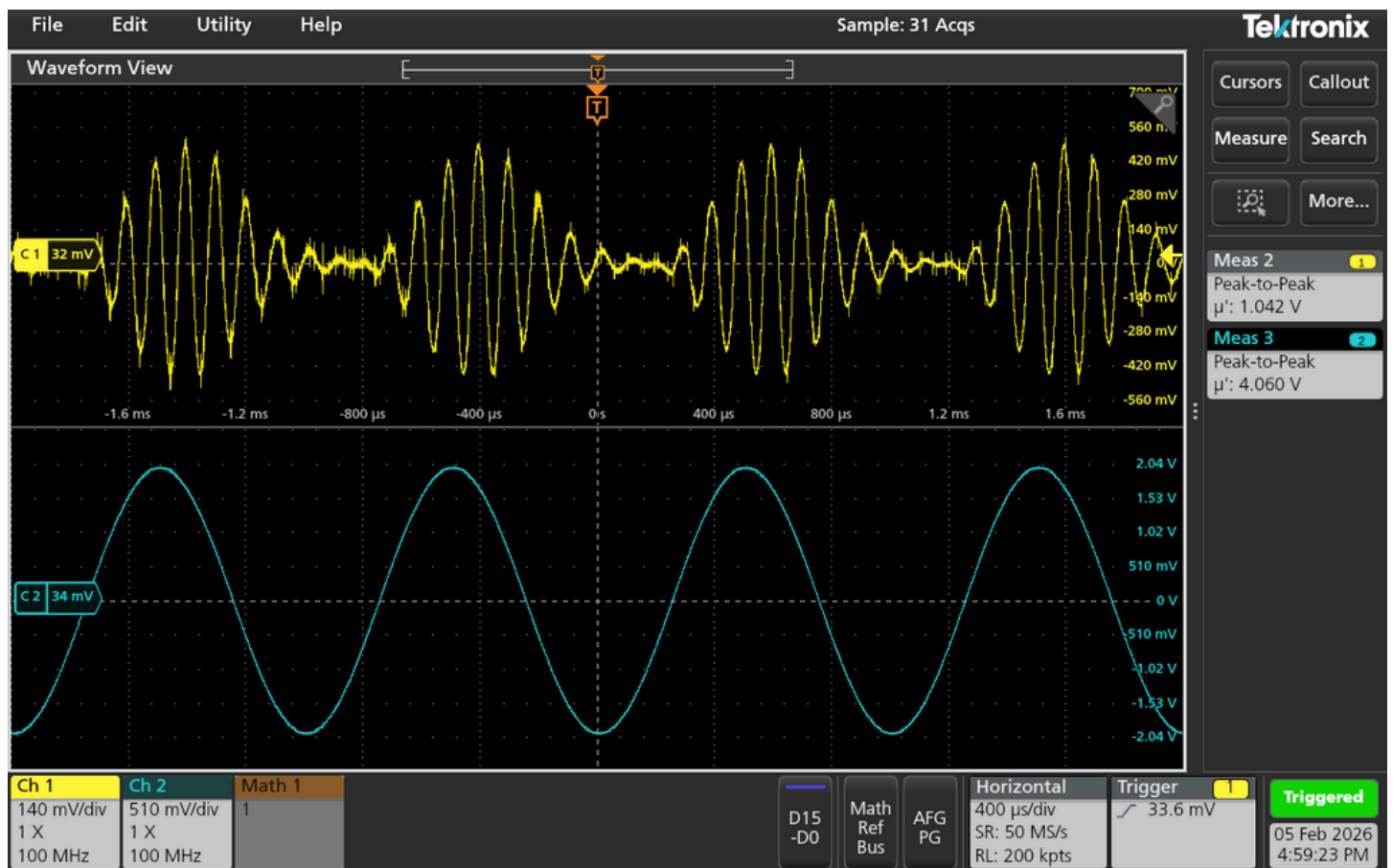


Figure 6: Amplitude modulates signal

Time-Domain Observation of the Amplitude modulated signal:

A clean amplitude-modulated waveform is obtained.

The carrier frequency component is suppressed.

No visible distortion in the 9–11 kHz band.

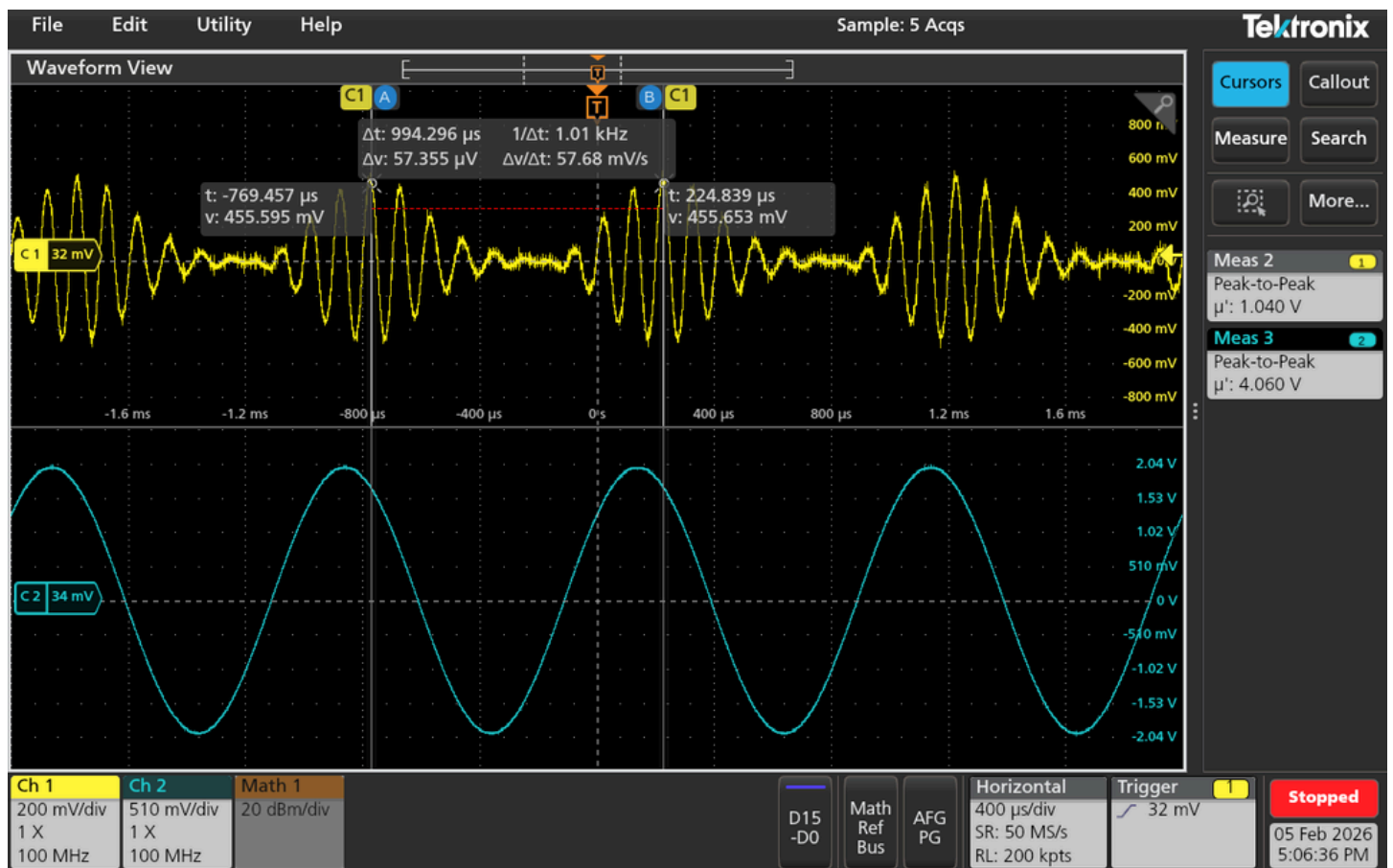


Figure 7: Calculation of f_m using time period

Message Signal

From CRO:

Measured time period of message signal:

$$F_m = 1/T_m = 1.01 \text{ kHz}$$

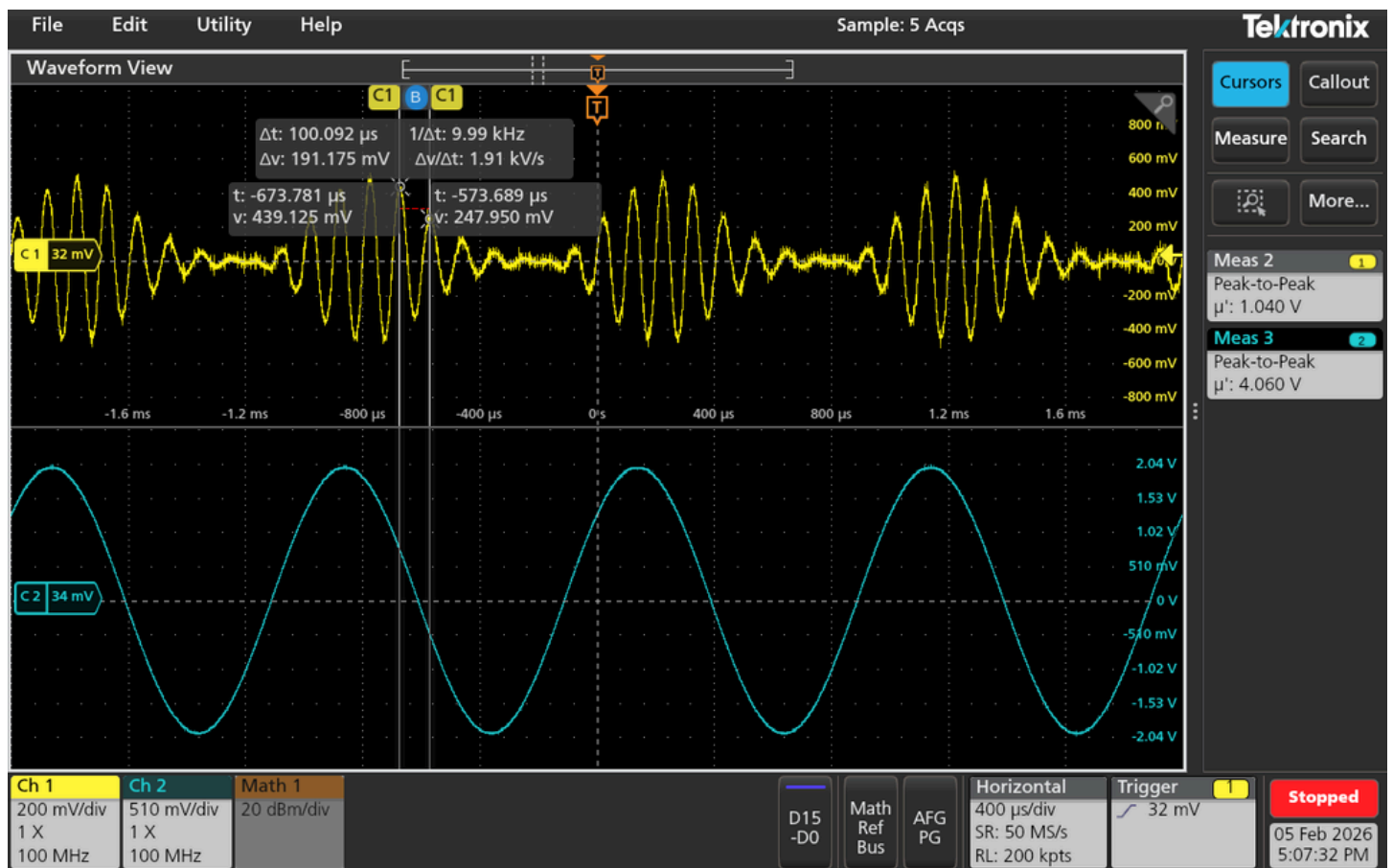


Figure 8: Calculation of f_c using time period

Carrier Signal

Measured time period of square wave carrier:

$$F_c = 1/T_c = 9.99 kHz$$

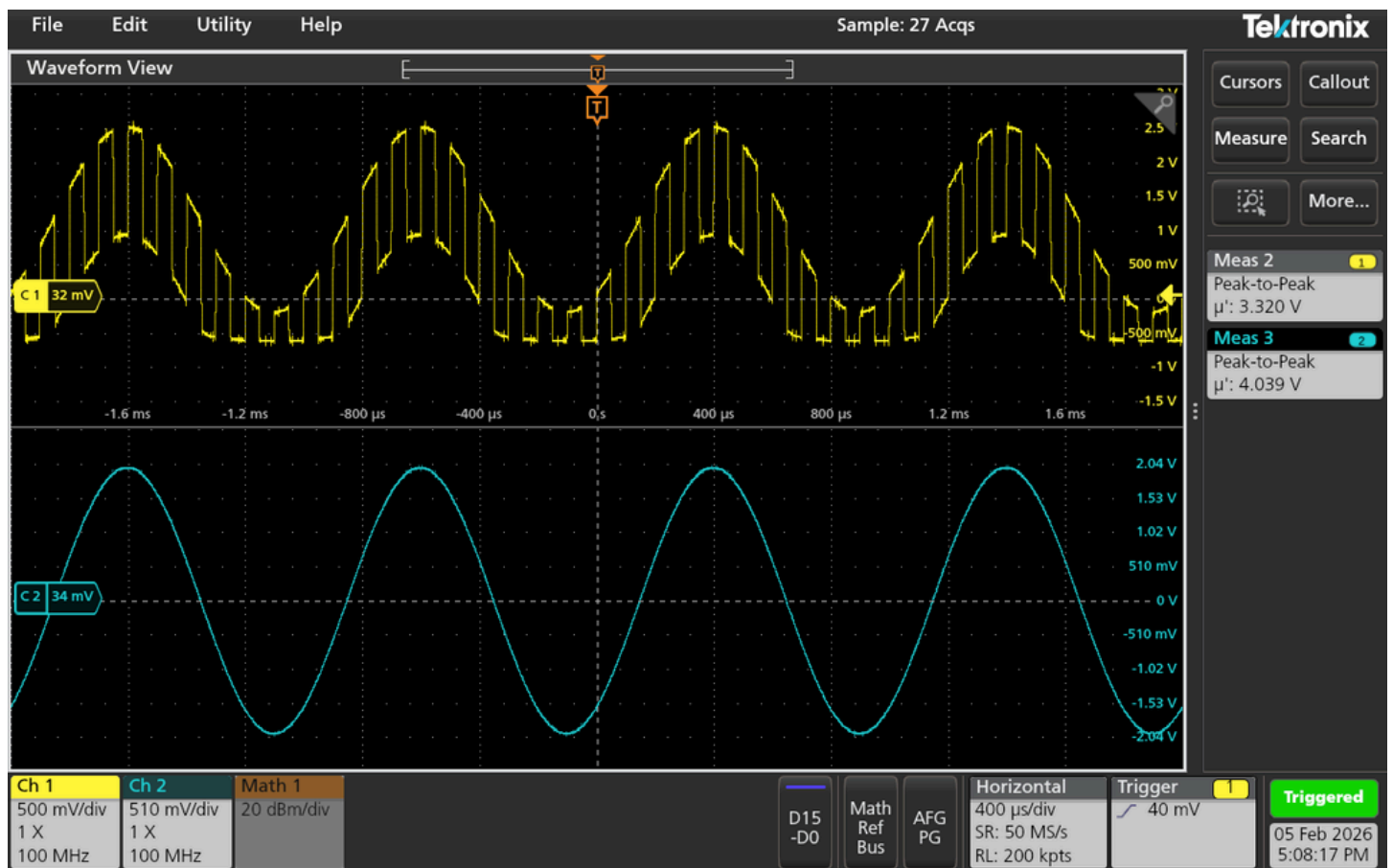


Figure 9: Case when carrier has no -2.5V DC offset



Figure 10: FFT spectrum of the DSB-SC signal showing dominant sidebands at 9 kHz and 11 kHz.

FFT Observation of AM Signal

Strong peaks observed at 9 kHz ($f_c - f_m$) and 11 kHz ($f_c + f_m$), confirms DSB-SC modulation.

Carrier at 10 kHz is significantly reduced (~ -5 dBm), indicating carrier suppression.

Additional smaller peaks at 30 kHz and 50 kHz due to odd harmonics of the square-wave carrier.

Small component at 1 kHz appears due to DC interaction.

Higher-order harmonics are strongly attenuated after the band-pass filter.

6. Discussion

- The switching modulator uses the BFW11 JFET as a high-speed switch to multiply the sinusoidal message with the square-wave carrier. Since a square wave contains only odd harmonics, the output spectrum consists of sidebands at $f_c + f_m$ and $f_c - f_m$, along with residual carrier and higher odd harmonics ($3f_c$, $5f_c$, etc.), as verified by the FFT.
- The carrier amplitude was kept much larger than the message amplitude to ensure proper switching and avoid clipping. A negative DC offset was applied to the carrier to maintain correct V_{gs} biasing. Without this offset, switching becomes asymmetric and carrier leakage increases. However, adding DC offset reduces power efficiency and may cause the carrier to become unsuppressed. When DC offset is applied, the coupling capacitor must be removed to prevent blocking of the DC component.
- A unity gain buffer was used between stages to minimize loading effects and improve signal isolation. The second-order multiple feedback Band-Pass Filter was necessary because a first-order filter has insufficient roll-off to suppress harmonics. The BPF effectively isolated the desired band, though minor harmonics remained due to finite roll-off.
- The LM741 op-amp's limited gain-bandwidth restricts high-frequency performance, making the setup unsuitable for high-quality industrial use. Minor distortions were caused by component tolerances, breadboard parasitics, and non-ideal switching. Care was taken to avoid positive gate voltage on the JFET to prevent device damage.
- Overall, the experiment demonstrates successful DSB-SC generation using switching modulation, while highlighting practical limitations such as harmonic generation, bias sensitivity, and power inefficiency.